

Pontificia Universidad Javeriana
Computación de alto desempeño
Laboratorio_ML00: Configuración Clúster SPARK

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1. Objetivo

Configuración un Clúster de Spark con VM Linux Rocky.

2. Desarrollo del Taller

A continuación, se documenta cada paso del proceso de configuración del clúster Spark. Para cada paso, se ejecutaron los comandos indicados y se registró la evidencia (captura de pantalla) en el espacio habilitado.

2A — SSH Passwordless (Master → Workers)

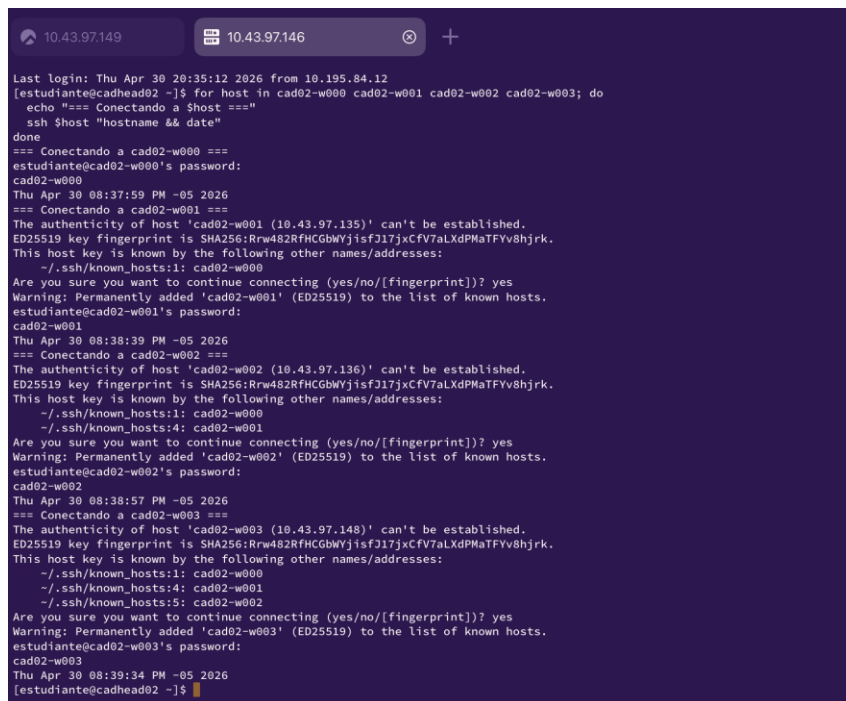
Desde el nodo master, se configuró el acceso SSH sin contraseña hacia todos los workers. Esto permitió que el master pueda iniciar procesos en los workers sin requerir autenticación manual.

Paso 1: Generar el par de claves SSH en el master:

```
$ ssh-keygen
```

Paso 2: Copiar la clave pública al master y a cada worker:

```
$ ssh-copy-id master@<ip-del-master>
$ ssh-copy-id worker@<ip-del-worker>
```

A terminal window with a dark background and light text. At the top, there are two tabs with IP addresses: 10.43.97.149 and 10.43.97.146. The terminal output shows the execution of a script that generates SSH keys and copies them to three worker nodes (cad02-w000, cad02-w001, and cad02-w002). The script uses 'ssh-keygen' to generate keys and 'ssh-copy-id' to copy the public key to each worker. The output shows the progress of this process, including prompts for confirmation and the addition of host fingerprints to the known hosts list. The script concludes with a final confirmation prompt for each worker node.

```
Last login: Thu Apr 30 20:35:12 2026 from 10.195.84.12
[estudiante@cadhead02 ~]$ for host in cad02-w000 cad02-w001 cad02-w002 cad02-w003; do
  echo "=== Conectando a $host ==="
  ssh $host "hostname && date"
done
=== Conectando a cad02-w000 ===
estudiante@cad02-w000's password:
cad02-w000
Thu Apr 30 08:37:59 PM -05 2026
=== Conectando a cad02-w001 ===
The authenticity of host 'cad02-w001 (10.43.97.135)' can't be established.
ED25519 key fingerprint is SHA256:Rrw482RfHCGbWvjisfJ17jxCfV7aLXdPMaTFVv8hjrK.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: cad02-w000
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'cad02-w001' (ED25519) to the list of known hosts.
estudiante@cad02-w001's password:
cad02-w001
Thu Apr 30 08:38:39 PM -05 2026
=== Conectando a cad02-w002 ===
The authenticity of host 'cad02-w002 (10.43.97.136)' can't be established.
ED25519 key fingerprint is SHA256:Rrw482RfHCGbWvjisfJ17jxCfV7aLXdPMaTFVv8hjrK.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: cad02-w000
  ~/.ssh/known_hosts:4: cad02-w001
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'cad02-w002' (ED25519) to the list of known hosts.
estudiante@cad02-w002's password:
cad02-w002
Thu Apr 30 08:38:57 PM -05 2026
=== Conectando a cad02-w003 ===
The authenticity of host 'cad02-w003 (10.43.97.148)' can't be established.
ED25519 key fingerprint is SHA256:Rrw482RfHCGbWvjisfJ17jxCfV7aLXdPMaTFVv8hjrK.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: cad02-w000
  ~/.ssh/known_hosts:4: cad02-w001
  ~/.ssh/known_hosts:5: cad02-w002
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'cad02-w003' (ED25519) to the list of known hosts.
estudiante@cad02-w003's password:
cad02-w003
Thu Apr 30 08:39:34 PM -05 2026
[estudiante@cadhead02 ~]$
```

2B — Edición del Fichero /etc/hosts

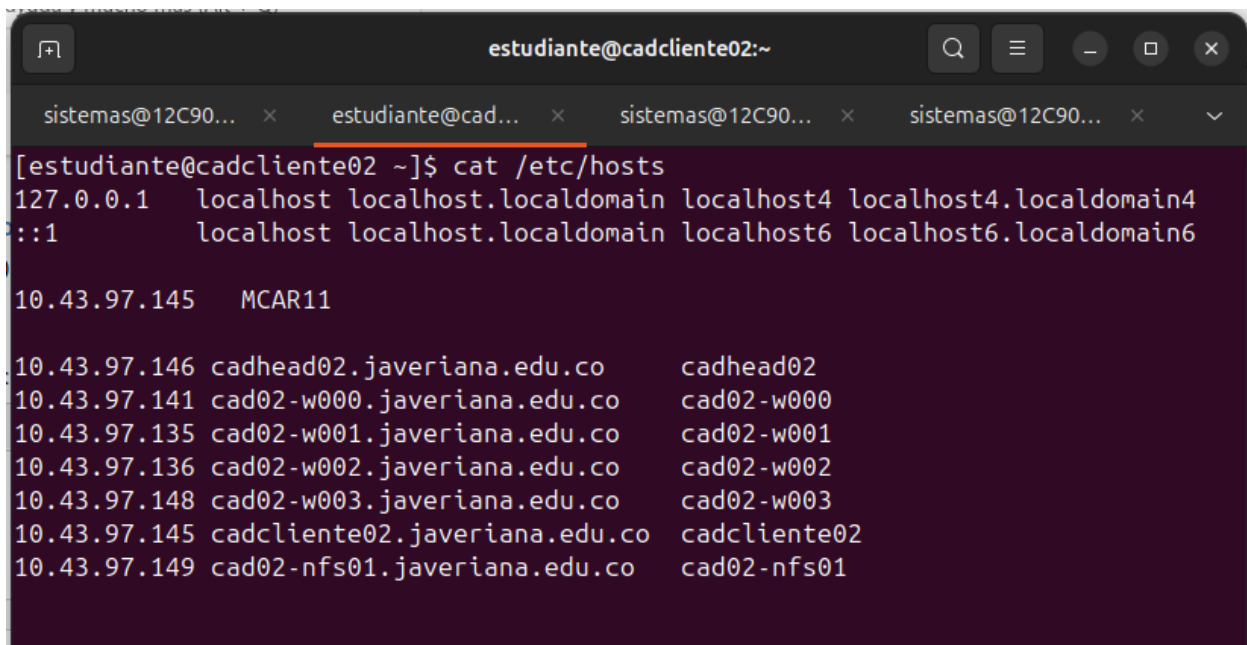
En TODOS los nodos (master y workers), se editó el fichero /etc/hosts para agregar la resolución de nombres de los nodos del clúster.

Paso 1: Abrir el fichero hosts con nano:

```
$ nano /etc/hosts
```

Paso 2: Agregar las siguientes líneas al fichero:

```
#####  
# IP's Master/Workers  
#####  
10.43.97.146 cadhead02.javeriana.edu.co cadhead02 10.43.97.141 cad02-  
w000.javeriana.edu.co cad02-w000 10.43.97.135 cad02-w001.javeriana.edu.co cad02-  
w001 10.43.97.136 cad02-w002.javeriana.edu.co cad02-w002 10.43.97.148 cad02-  
w003.javeriana.edu.co cad02-w003 10.43.97.145 cadcliente02.javeriana.edu.co  
cadcliente02 10.43.97.149 cad02-nfs01.javeriana.edu.co cad02-nfs01
```



The screenshot shows a terminal window titled 'estudiante@cadcliente02:~'. The terminal displays the output of the command 'cat /etc/hosts'. The output lists several IP addresses and their corresponding hostnames, including localhost, cadhead02, cad02-w000, cad02-w001, cad02-w002, cad02-w003, cadcliente02, and cad02-nfs01. The terminal window has a dark background and a light-colored text.

```
estudiante@cadcliente02:~  
[estudiante@cadcliente02 ~]$ cat /etc/hosts  
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4  
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6  
  
10.43.97.145  MCAR11  
  
10.43.97.146 cadhead02.javeriana.edu.co    cadhead02  
10.43.97.141 cad02-w000.javeriana.edu.co    cad02-w000  
10.43.97.135 cad02-w001.javeriana.edu.co    cad02-w001  
10.43.97.136 cad02-w002.javeriana.edu.co    cad02-w002  
10.43.97.148 cad02-w003.javeriana.edu.co    cad02-w003  
10.43.97.145 cadcliente02.javeriana.edu.co  cadcliente02  
10.43.97.149 cad02-nfs01.javeriana.edu.co    cad02-nfs01
```

2C — Creación de la Carpeta Compartida en el Master

En el nodo master, se creó el directorio /nfs/condor que fue compartido vía NFS con los workers.

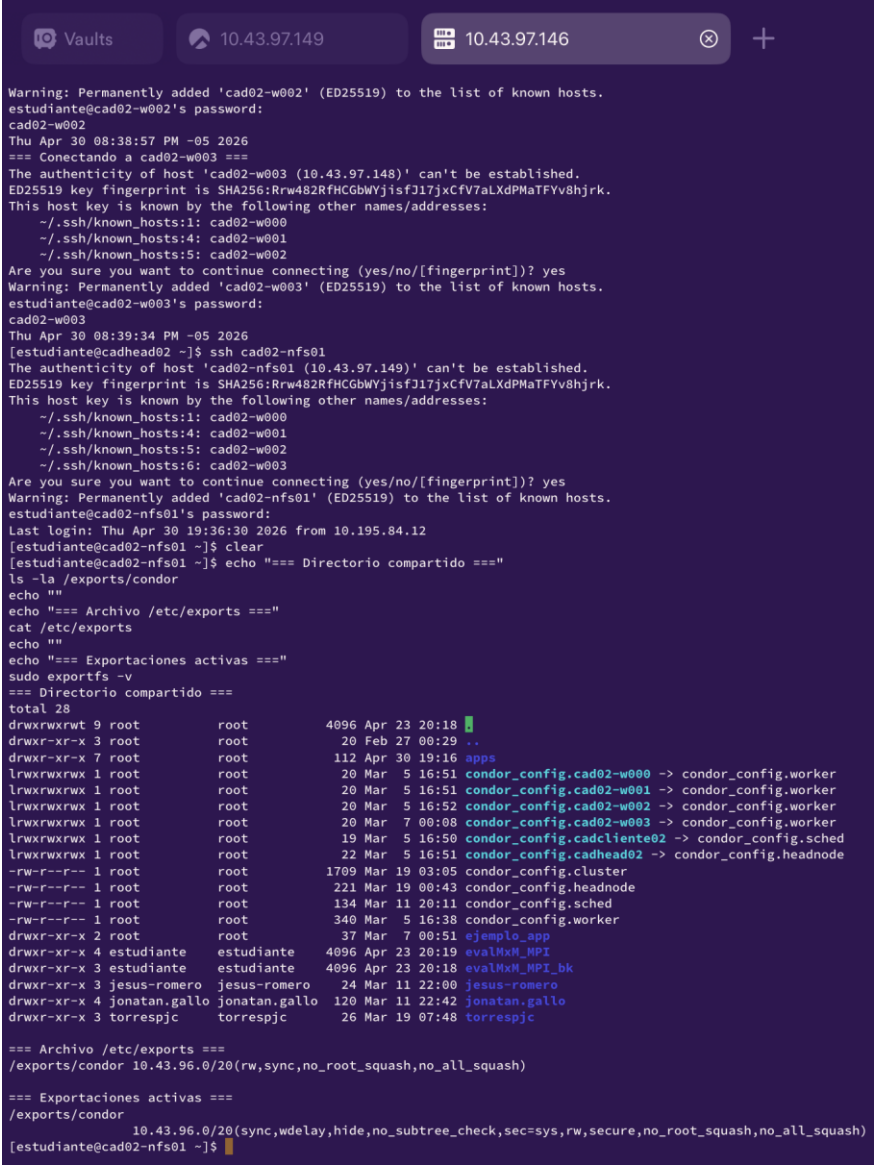
Paso 1: Crear el directorio /nfs/condor:

```
$ cd /
$ sudo mkdir /nfs/condor -p
$ sudo chmod estudiante:estudiante nfs/condor
```

Paso 2: Editar /etc/exports para compartir la carpeta:

```
$ sudo nano /etc/exports

# Agregar la siguiente línea:
```



The screenshot shows a terminal window with a dark background. At the top, there are three tabs: 'Vaults', '10.43.97.149', and '10.43.97.146'. The terminal output shows a series of SSH connections and file system operations. It starts with a warning about adding 'cad02-w002' to the list of known hosts, followed by a successful connection to 'cad02-w003'. Then, it shows a connection to 'cad02-nfs01' with a warning about adding it to the list of known hosts. The user then runs 'clear', 'ls -la /exports/condor', and 'cat /etc/exports'. The output of 'cat /etc/exports' shows the current configuration, including the 'condor' directory being exported to various hosts. The user then runs 'sudo exportfs -v' to show the active exports, which lists the 'condor' directory being exported to '10.43.96.0/20' and '10.43.96.0/20' with various options. The terminal ends with the user running '[estudiante@cad02-nfs01 ~]\$'.

```
Warning: Permanently added 'cad02-w002' (ED25519) to the list of known hosts.
estudiante@cad02-w002's password:
cad02-w002
Thu Apr 30 08:38:57 PM -05 2026
=== Conectando a cad02-w003 ===
The authenticity of host 'cad02-w003 (10.43.97.148)' can't be established.
ED25519 key fingerprint is SHA256:Rrw482RfHCgbWYjisfJ17jxCfV7aLXdPMaTFYv8hjrK.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: cad02-w000
  ~/.ssh/known_hosts:4: cad02-w001
  ~/.ssh/known_hosts:5: cad02-w002
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'cad02-w003' (ED25519) to the list of known hosts.
estudiante@cad02-w003's password:
cad02-w003
Thu Apr 30 08:39:34 PM -05 2026
[estudiante@cadhead02 ~]$ ssh cad02-nfs01
The authenticity of host 'cad02-nfs01 (10.43.97.149)' can't be established.
ED25519 key fingerprint is SHA256:Rrw482RfHCgbWYjisfJ17jxCfV7aLXdPMaTFYv8hjrK.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: cad02-w000
  ~/.ssh/known_hosts:4: cad02-w001
  ~/.ssh/known_hosts:5: cad02-w002
  ~/.ssh/known_hosts:6: cad02-w003
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'cad02-nfs01' (ED25519) to the list of known hosts.
estudiante@cad02-nfs01's password:
Last login: Thu Apr 30 10:36:30 2026 from 10.195.84.12
[estudiante@cad02-nfs01 ~]$ clear
[estudiante@cad02-nfs01 ~]$ echo "=== Directorio compartido ==="
ls -la /exports/condor
echo ""
echo "=== Archivo /etc/exports ==="
cat /etc/exports
echo ""
echo "=== Exportaciones activas ==="
sudo exportfs -v
=== Directorio compartido ===
total 28
drwxrwxrwt 9 root      root      4096 Apr 23 20:18 .
drwxr-xr-x 3 root      root      20 Feb 27 00:29 ..
drwxr-xr-x 7 root      root      112 Apr 30 19:16 apps
lrwxrwxrwx 1 root      root         0 Mar  5 16:51 condor_config.cad02-w000 -> condor_config.worker
lrwxrwxrwx 1 root      root         0 Mar  5 16:51 condor_config.cad02-w001 -> condor_config.worker
lrwxrwxrwx 1 root      root         0 Mar  5 16:52 condor_config.cad02-w002 -> condor_config.worker
lrwxrwxrwx 1 root      root         0 Mar  7 00:08 condor_config.cad02-w003 -> condor_config.worker
lrwxrwxrwx 1 root      root         0 Mar  5 16:50 condor_config.cadcliente02 -> condor_config.sched
lrwxrwxrwx 1 root      root         0 Mar  5 16:51 condor_config.cadhead02 -> condor_config.headnode
-rw-r--r-- 1 root      root         0 1709 Mar 19 03:05 condor_config.cluster
-rw-r--r-- 1 root      root         0 221 Mar 19 00:43 condor_config.headnode
-rw-r--r-- 1 root      root         0 134 Mar 11 20:11 condor_config.sched
-rw-r--r-- 1 root      root         0 340 Mar  5 16:38 condor_config.worker
drwxr-xr-x 2 root      root         0 37 Mar  7 00:51 ejemplo_app
drwxr-xr-x 4 estudiante estudiante 4096 Apr 23 20:19 evalMxM_MPI
drwxr-xr-x 3 estudiante estudiante 4096 Apr 23 20:18 evalMxM_MPI_bk
drwxr-xr-x 3 jesus-romero jesus-romero 24 Mar 11 22:00 jesus-romero
drwxr-xr-x 4 jonatan.gallo jonatan.gallo 120 Mar 11 22:42 jonatan.gallo
drwxr-xr-x 3 torrespjc torrespjc 26 Mar 19 07:48 torrespjc

=== Archivo /etc/exports ===
/exports/condor 10.43.96.0/20(rw,sync,no_root_squash,no_all_squash)

=== Exportaciones activas ===
/exports/condor
10.43.96.0/20(sync,wdelay,hide,no_subtree_check,sec=sys,rw,secure,no_root_squash,no_all_squash)
[estudiante@cad02-nfs01 ~]$
```

```
/nfs/condor <ip_cliente>(rw,no_root_squash,subtree_check,fsid=0)
```

2D — Instalación y Configuración del Servicio NFS

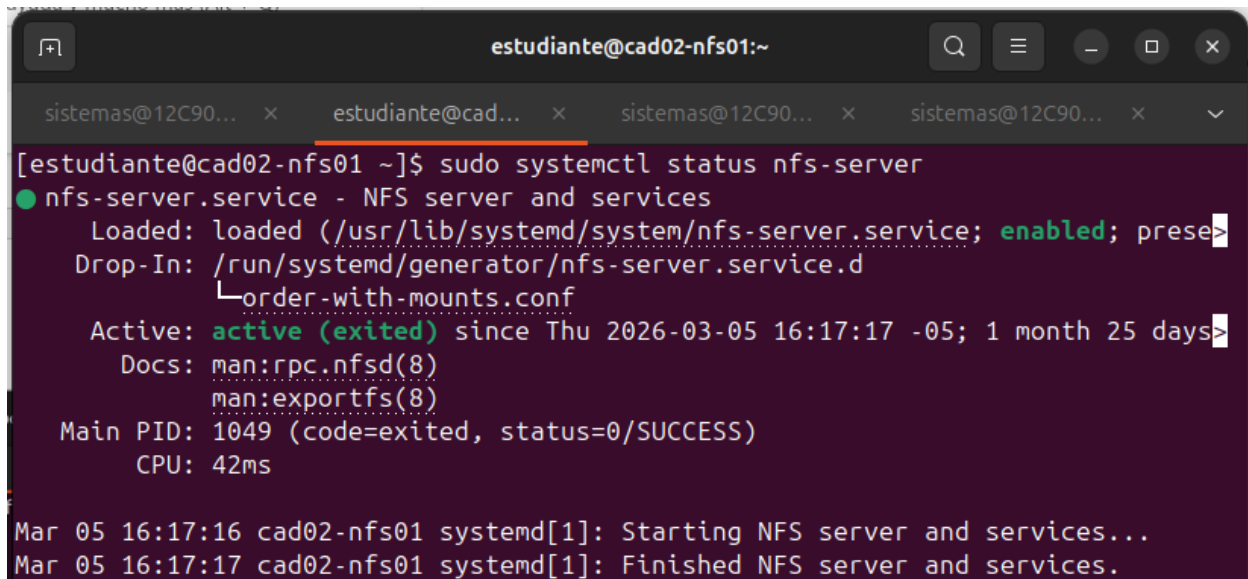
Se instaló y activó el servicio NFS en el nodo master para permitir el montaje de la carpeta compartida desde los workers.

Paso 1: Instalar las utilidades NFS:

```
$ sudo dnf install nfs-utils
```

Paso 2: Habilitar, iniciar y verificar el servicio:

```
$ sudo systemctl enable nfs-server
$ sudo systemctl start nfs-server
$ sudo systemctl status nfs-server
```



```
estudiante@cad02-nfs01:~
sistemas@12C90... x estudiante@cad... x sistemas@12C90... x sistemas@12C90... x
[estudiante@cad02-nfs01 ~]$ sudo systemctl status nfs-server
● nfs-server.service - NFS server and services
   Loaded: loaded (/usr/lib/systemd/system/nfs-server.service; enabled; prese>
   Drop-In: /run/systemd/generator/nfs-server.service.d
            └─order-with-mounts.conf
   Active: active (exited) since Thu 2026-03-05 16:17:17 -05; 1 month 25 days>
   Docs: man:rpc.nfsd(8)
         man:exportfs(8)
   Main PID: 1049 (code=exited, status=0/SUCCESS)
   CPU: 42ms

Mar 05 16:17:16 cad02-nfs01 systemd[1]: Starting NFS server and services...
Mar 05 16:17:17 cad02-nfs01 systemd[1]: Finished NFS server and services.
```

2E — Montaje del NFS en los Workers

En cada nodo worker, se creó el directorio de montaje y se conectó la carpeta compartida del master.

Paso 1: Crear directorio de montaje en el worker:

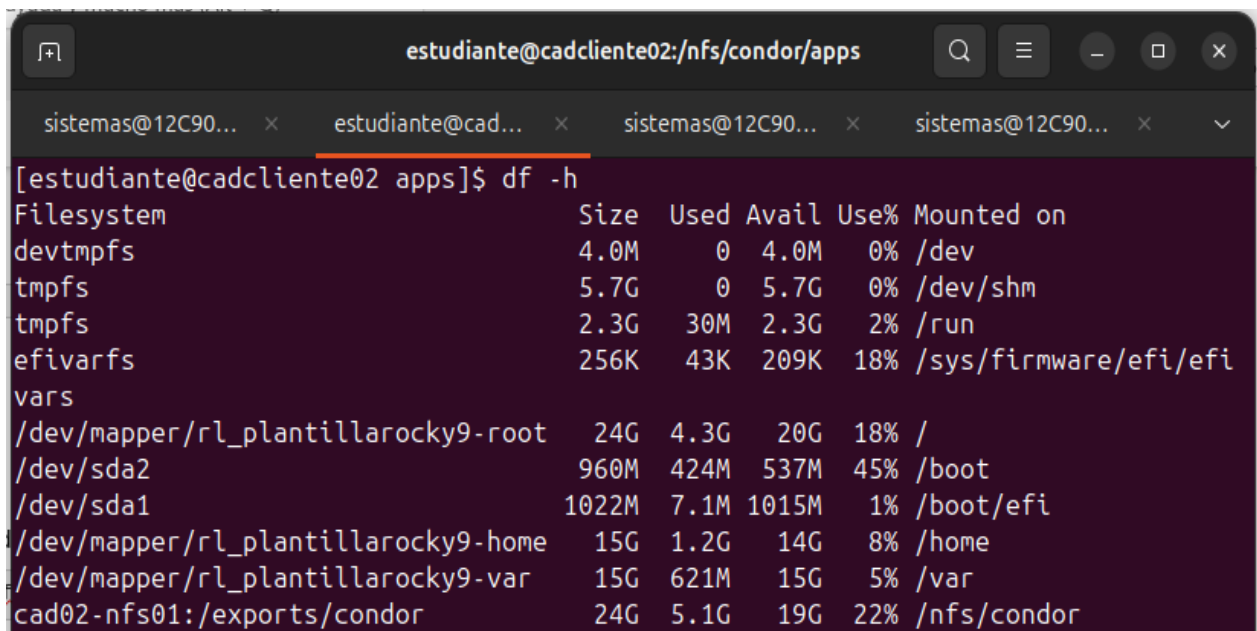
```
$ cd /  
$ sudo mkdir -p nfs/condor
```

Paso 2: Montar la carpeta compartida del master:

```
$ mount 10.43.97.149:/nfs/condor /nfs/condor
```

Paso 3: Verificar los sistemas de ficheros montados:

```
$ df -h
```



```
estudiante@cadcliente02:/nfs/condor/apps  
[estudiante@cadcliente02 apps]$ df -h  
Filesystem      Size  Used Avail Use% Mounted on  
devtmpfs        4.0M   0  4.0M   0% /dev  
tmpfs           5.7G   0  5.7G   0% /dev/shm  
tmpfs           2.3G  30M  2.3G   2% /run  
efivarfs        256K  43K  209K  18% /sys/firmware/efi/efi  
vars  
/dev/mapper/rl_plantillarocky9-root 24G  4.3G   20G  18% /  
/dev/sda2       960M  424M  537M  45% /boot  
/dev/sda1      1022M   7.1M 1015M   1% /boot/efi  
/dev/mapper/rl_plantillarocky9-home 15G  1.2G   14G   8% /home  
/dev/mapper/rl_plantillarocky9-var  15G  621M   15G   5% /var  
cad02-nfs01:/exports/condor        24G  5.1G   19G  22% /nfs/condor
```

2F — Instalación de Apache Spark en la Carpeta Compartida

Desde el master, se copió el paquete de Spark a la carpeta compartida y se descomprimió. Esto permitió que todos los nodos accedan a los mismos binarios vía NFS.

Paso 1: Copiar el archivo .tgz de Spark a /nfs/condor/apps:

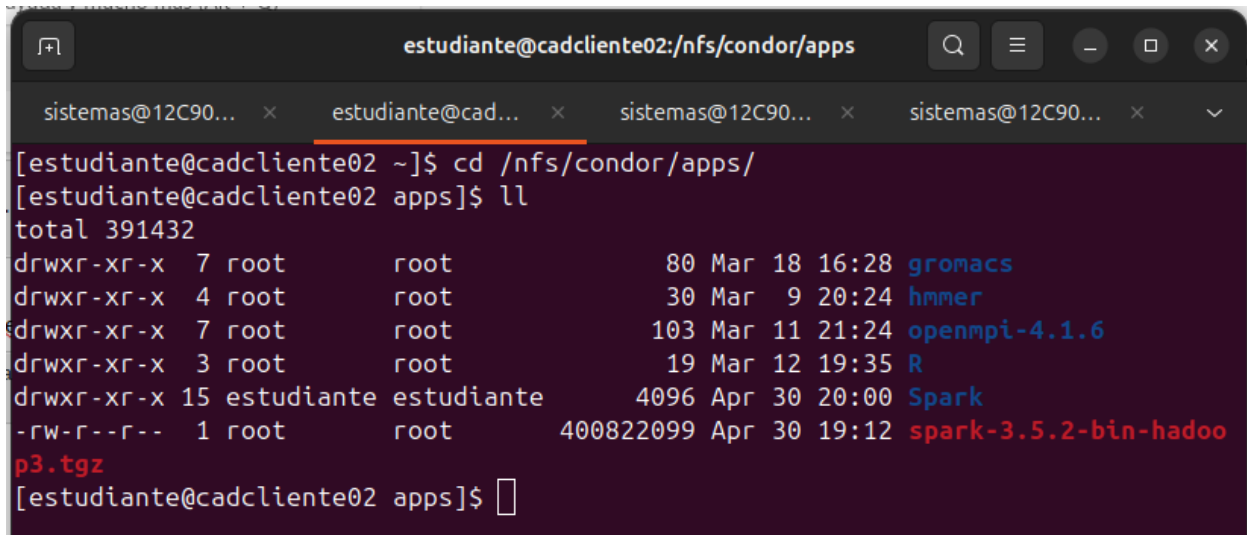
```
$ cp /home/estudiante/spark-3.5.2-bin-hadoop3.tgz /nfs/condor/apps
```

Paso 2: Descomprimir y renombrar la carpeta:

```
$ cd /nfs/condor/apps
$ tar -zxvf spark-3.5.2-bin-hadoop3.tgz
$ mv spark-3.5.2-bin-hadoop3 Spark
```

Paso 3: Instalar Java en master y workers:

```
$ sudo dnf install openjdk-8-jdk-headless
$ java -version
```

A terminal window titled 'estudiante@cadcliente02:/nfs/condor/apps' showing the contents of the directory. The user has run 'cd /nfs/condor/apps/' and 'll'. The output shows a list of files and directories with permissions, owner, group, size, date, and name. The files listed are gromacs, hammer, openmpi-4.1.6, R, Spark, and spark-3.5.2-bin-hadoop3.tgz. The terminal has a dark background with light-colored text.

```
estudiante@cadcliente02:/nfs/condor/apps
[estudiante@cadcliente02 ~]$ cd /nfs/condor/apps/
[estudiante@cadcliente02 apps]$ ll
total 391432
drwxr-xr-x  7 root      root          80 Mar 18 16:28 gromacs
drwxr-xr-x  4 root      root          30 Mar  9 20:24 hammer
drwxr-xr-x  7 root      root        103 Mar 11 21:24 openmpi-4.1.6
drwxr-xr-x  3 root      root          19 Mar 12 19:35 R
drwxr-xr-x 15 estudiante estudiante 4096 Apr 30 20:00 Spark
-rw-r--r--  1 root      root    400822099 Apr 30 19:12 spark-3.5.2-bin-hadoo
p3.tgz
[estudiante@cadcliente02 apps]$
```

2G — Configuración de Variables de Entorno (Todos los Nodos)

En TODOS los nodos (master y workers), se configuró las variables de entorno para Spark y Python editando el fichero `.bashrc`.

Paso 1: Verificar el estado actual del entorno Spark:

```
$ env | grep -i SPARK
```

Paso 2: Editar el fichero `.bashrc`:

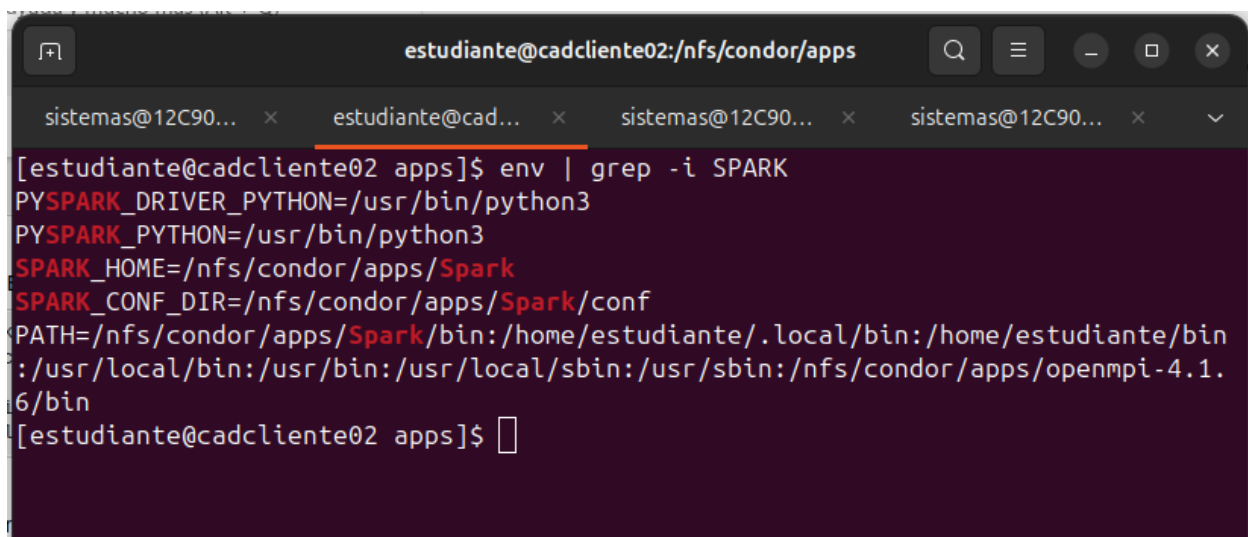
```
$ cd  
$ nano .bashrc
```

Paso 3: Buscar la línea `# export SYSTEMD_PAGER=` e insertar después:

```
export SPARK_HOME="/nfs/condor/Spark"  
export SPARK_CONF_DIR="$SPARK_HOME/conf"  
export PATH="$SPARK_HOME/bin":$PATH  
export PYSPARK_PYTHON="/usr/local/bin/python3"  
export PYSPARK_DRIVER_PYTHON="/usr/local/bin/python3"
```

Paso 4: Recargar la configuración y verificar:

```
$ source .bashrc  
$ env | grep -i SPARK
```



The screenshot shows a terminal window with the title bar 'estudiante@cadcliente02:/nfs/condor/apps'. The terminal output is as follows:

```
[estudiante@cadcliente02 apps]$ env | grep -i SPARK  
PYSPARK_DRIVER_PYTHON=/usr/bin/python3  
PYSPARK_PYTHON=/usr/bin/python3  
SPARK_HOME=/nfs/condor/apps/Spark  
SPARK_CONF_DIR=/nfs/condor/apps/Spark/conf  
PATH=/nfs/condor/apps/Spark/bin:/home/estudiante/.local/bin:/home/estudiante/bin  
:/usr/local/bin:/usr/bin:/usr/local/sbin:/usr/sbin:/nfs/condor/apps/openmpi-4.1.  
6/bin  
[estudiante@cadcliente02 apps]$
```


2H — Configuración de Apache Spark

Se configuró los ficheros de entorno de Spark: spark-env.sh (dirección IP del master) y workers (lista de nodos worker).

Paso 1: Ir al directorio conf y crear el fichero de configuración:

```
$ cd /nfs/condor/apps/Spark/conf
$ cp spark-env.sh.template spark-env.sh
```

Paso 2: Editar spark-env.sh — buscar la sección de daemons e insertar la IP del master:

```
export SPARK_MASTER_HOST=10.43.97.145
export SPARK_WORKER_PORT=8888
export SPARK_WORKER_WEBUI_PORT=8081

export SPARK_DAEMON_MEMORY=1g

export SPARK_WORKER_CORES=4
export SPARK_WORKER_MEMORY=12g
export SPARK_DAEMON_JAVA_OPTS="-Dspark.schedules.mode=FAIR"

export SPARK_EXECUTOR_CORES=2
export SPARK_EXECUTOR_MEMORY=2g

export PYSARK_PYTHON=/usr/bin/python3
export PYSARK_DRIVER_PYTHON=/usr/bin/python3
```

Paso 3: Crear y editar el fichero workers:

```
# A Spark Worker will be started on each of the machines listed below.
# localhost
cad02-w000
cad02-w001
cad02-w002
cad02-w003
cadhead02
```

Paso 4: Instalar herramienta para ver procesos Java:

```
$ sudo dnf -y install java-1.8.0-openjdk-devel
$ jps
```

```
for HOST in 10.43.97.141 10.43.97.135 10.43.97.136 10.43.97.148 10.43.97.146 10.43.97.145
do
    echo "Installing $HOST..."
    ssh estudiante@$HOST "sudo dnf install -y java-1.8.0-openjdk-devel"
done
```

2I — Inicio del Servicio Spark y Verificación

Se levantó el clúster Spark y verifique que el master y los workers estén activos tanto por consola como por la interfaz web.

Paso 1: Iniciar todos los servicios Spark desde el master:

```
$ cd /nfs/condor/Spark/sbin
$ ./start-all.sh
```

Paso 2: Verificar los procesos Java activos (debe aparecer Master y Worker):

```
$ jps
```

Paso 3: Verificar el clúster en el navegador (reemplazar nn con el número de IP):

```
http://10.43.103.45:8080
```

Spark Master at spark://cadcliente02.javeriana.edu.co:7077

No es seguro 10.43.97.145:8080

spark 3.5.2

Spark Master at spark://cadcliente02.javeriana.edu.co:7077

URL: spark://cadcliente02.javeriana.edu.co:7077

Alive Workers: 5

Cores in use: 20 Total, 0 Used

Memory in use: 60.0 GiB Total, 0.0 B Used

Resources in use:

Applications: 0 Running, 0 Completed

Drivers: 0 Running, 0 Completed

Status: ALIVE

Workers (5)

Worker Id	Address	State	Cores	Memory	Resources
worker-20260430200203-10.43.97.136-8888	10.43.97.136:8888	ALIVE	4 (0 Used)	12.0 GiB (0.0 B Used)	
worker-20260430200203-10.43.97.141-8888	10.43.97.141:8888	ALIVE	4 (0 Used)	12.0 GiB (0.0 B Used)	
worker-20260430200204-10.43.97.135-8888	10.43.97.135:8888	ALIVE	4 (0 Used)	12.0 GiB (0.0 B Used)	
worker-20260430200204-10.43.97.146-8888	10.43.97.146:8888	ALIVE	4 (0 Used)	12.0 GiB (0.0 B Used)	
worker-20260430200204-10.43.97.148-8888	10.43.97.148:8888	ALIVE	4 (0 Used)	12.0 GiB (0.0 B Used)	

Running Applications (0)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
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Completed Applications (0)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
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3. Conclusiones

La configuración exitosa de un clúster Apache Spark sobre VMs con Rocky Linux permitió consolidar conocimientos fundamentales en computación de alto desempeño distribuida. A continuación, se presentan las principales conclusiones del laboratorio:

- La configuración de SSH sin contraseña entre el nodo master y los workers es un requisito previo indispensable para el funcionamiento del clúster, ya que Spark depende de este mecanismo para iniciar y coordinar procesos remotos de forma automática durante el lanzamiento del servicio mediante `start-all.sh`.
- El uso de NFS como sistema de archivos compartido demostró ser una estrategia eficiente para gestionar los binarios de Spark en un entorno multi-nodo: al instalar Spark una sola vez en `/nfs/condor/apps` y montarlo en todos los workers, se garantiza consistencia en versiones y se simplifica significativamente el mantenimiento del clúster.
- La correcta resolución de nombres mediante `/etc/hosts` en todos los nodos resultó crítica para la comunicación interna del clúster, evitando dependencias de un servidor DNS externo y asegurando que el master pueda identificar y contactar a cada worker por nombre de host.
- La configuración del archivo `spark-env.sh` permitió ajustar parámetros clave del clúster como la memoria por worker (12g), los cores disponibles (4 por nodo) y el modo de planificación FAIR, lo que demuestra la flexibilidad de Spark para adaptarse a los recursos físicos disponibles en la infraestructura.
- Finalmente, la verificación a través de la interfaz web del Spark Master (<http://10.43.97.145:8080>) confirmó que los 5 workers quedaron en estado **ALIVE** con un total de 20 cores y 60 GB de memoria disponibles para procesamiento distribuido, validando que el clúster quedó operativo y listo para ejecutar cargas de trabajo de machine learning y análisis de datos a gran escala.